

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

Category	Discipline Specific Core -III			
Course Title	Digital Forensics & Incident Response (DFIR)			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	3	0	4	5
Pre-requisites (if any)	• CSF & NS (Subject 1) completed or concurrent – log analysis and Splunk introduced there carry directly into this subject • Basic computer hardware and file system knowledge • Lab machines: FTK Imager 4.x + Autopsy 4.x + Volatility 3 + Wireshark + Splunk Free pre-installed; pre-seeded forensic disk images loaded			

1. Course Objectives:

CO1	Understand digital forensic principles, investigation process, Locard's Exchange Principle, chain-of-custody requirements, and the legal framework (IT Act, DPDP Act 2023).
CO2	Apply FTK Imager and Autopsy to acquire forensic images, verify integrity, recover deleted files, and build an activity timeline.
CO3	Conduct Windows forensic analysis using Autopsy and Registry Explorer – extract artefacts from registry hives, event logs, prefetch files, browser history, and USB connection history.
CO4	Analyse network traffic using Wireshark and correlate security events using Splunk to detect attacks and reconstruct timelines.
CO5	Execute the NIST IR lifecycle and produce a professional forensic investigation report that is legally defensible and meets chain-of-custody standards.

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

2. Course Contents:

Unit	Course Content	Weightage	Hrs.
Unit I	Definition, scope, and types of digital forensics — disk, memory, network, mobile, cloud forensics, email forensics; goals of a forensic investigation; digital evidence concept, what counts as evidence; Locard's Exchange Principle in the digital context; the digital forensics investigation process (Identification, Preservation, Collection, Examination, Analysis, Reporting); chain of custody (why it matters, how to maintain it; case study walkthrough; legal and ethical aspects) IT Act 2000 (relevant sections), Indian Evidence Act provisions for electronic records, admissibility standards (Daubert/Frye overview); investigator roles, responsibilities, documentation best practices; common pitfalls (contamination of evidence, tool validation, anti-forensics awareness); introduction to forensic operating systems (CAINE, SIFT) for OS identification.	20%	7
Unit II	Volatile vs non-volatile data (what to acquire first and why); order of volatility; live vs dead acquisition; write blockers (hardware vs software) and why they are essential; imaging tools overview (FTK Imager, dd, EnCase, Autopsy; bit-stream copy concept); hash verification (MD5, SHA-1, SHA-256) for evidence integrity; case studies; chain of custody fields during imaging; deleted file recovery (file carving fundamentals, slack space, unallocated space; password recovery and bypass techniques (overview only ethical context); file system signatures; RAM/memory acquisition basics — why memory matters, tools (FTK Imager, Belkasoft RAM Capturer); evidence storage and handling.	20%	7
Unit III	Windows file system internals (FAT vs NTFS overview), MFT (Master File Table), \$MFT, \$LogFile, deleted file recovery from NTFS; Windows artifacts [Registry hives (SAM, SECURITY, SOFTWARE, SYSTEM, NTUSER.DAT)]; registry keys for USB device history, recently opened files, installed software; Event Logs (Security, System, Application); browser artifacts (history, cache, cookies, downloads); prefetch files; LNK files and shellbags; timeline analysis with log2timeline/Plaso; Linux forensics — ext3/ext4 file system overview, key directories (/var/log, /etc, /home, /tmp); critical log files (auth.log, syslog, bash_history); user account artifacts; cron jobs as persistence;	20%	8

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

	tool overview — Autopsy, Sleuth Kit, FTK; hands-on case scenario walkthrough.		
Unit IV	Network forensics fundamentals (what to capture, where to capture (TAPs, SPAN ports)); packet capture analysis with Wireshark for forensic purposes; NetFlow and traffic flow analysis; firewall, proxy, and IDS log analysis for incident reconstruction; identifying C2 traffic, data exfiltration patterns, and lateral movement; email forensics (header analysis, tracing the source, identifying spoofing); cloud forensics challenges - shared responsibility model, jurisdiction issues, evidence acquisition from AWS/Azure/GCP at a conceptual level; dark web overview and challenges for investigators; encryption challenges in forensics; mobile forensics introduction - iOS vs Android acquisition methods, common artifacts.	20%	7
Unit V	Incident Response lifecycle - NIST SP 800-61 phases (Preparation, Detection & Analysis, Containment, Eradication, Recovery, Post-Incident Activity); how forensics integrates with IR; first responder responsibilities; preserving volatile data during a live incident; malware forensics introduction - static vs dynamic analysis overview, sandbox tools (Cuckoo, Any.Run, Flare VM); ransomware case study; report writing - structure of a forensic report (executive summary, methodology, findings, evidence, conclusions); writing for technical and non-technical audiences (court, management); presenting evidence - affidavits, expert witness testimony basics; case studies - significant Indian and international cybercrime cases; emerging trends - AI in forensics, anti-forensic techniques, IoT forensics overview.	20%	7

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

Program Outcomes (POs)

PO Code	Program Outcomes
PO1	Computational Knowledge
PO2	Problem Analysis
PO3	Design and Development of Solutions
PO4	Modern Tool Usage
PO5	Individual and Team Work
PO6	Professional Ethics
PO7	Communication Skills
PO8	Life-long Learning

CO-PO Mapping Matrix

Mapping Scale:

- 3 = High
- 2 = Moderate
- 1 = Low

COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	2	2	2	3	2	2
CO2.	2	3	3	2	2	3	1	2
CO3	2	3	3	2	2	2	1	2
CO4	2	3	3	3	2	2	1	2
CO5	2	2	2	2	3	3	3	3

L J INSTITUTE OF COMPUTER APPLICATIONS

Master of Computer Applications (MCA)

Syllabus for Batch - 2025

Semester: III

List of Practicals (Samples)

Lab: FTK Imager, Autopsy, Volatility 3, Wireshark, Splunk Free installed. Pre-seeded forensic images and faculty-provided PCAP file loaded on lab machines.

Practical 1 – Evidence Acquisition with FTK Imager (Est. time: 3 hrs)

Objective: Capture volatile memory and create a forensic image of a USB device; verify integrity with hash values; complete chain-of-custody form.

- Environment: Windows lab machine; FTK Imager 4.x; instructor-provided USB drive
- Tools: FTK Imager
- Expected output: FTK Imager verification report PDF with matching hashes; screenshot of chain-of-custody fields filled in; RAM capture file size recorded.
- Observation questions: (1) You accidentally boot from the suspect USB drive before imaging it. Which files have their timestamps changed? What data might be written to the drive? (2) Why must you record MD5 AND SHA-1 – why is one hash algorithm not sufficient in court?

Practical 2 – Autopsy: Deleted File Recovery & Activity Timeline (Est. time: 4 hrs)

Objective: Load a pre-seeded forensic image in Autopsy; recover deleted files; extract recent activity artefacts; build a chronological timeline.

- Environment: Lab machine with Autopsy 4.x; faculty-provided Windows forensic .E01 image (contains deleted documents, browser history, planted evidence)
- Tools: Autopsy 4.x
- Expected output: Deleted file recovery screenshot; recent documents list; browser history screenshot; USB device table; timeline CSV; 5-event narrative.
- Observation questions: (1) In NTFS, a file is deleted. The MFT entry is marked inactive. When does the actual data become unrecoverable? (2) The timeline shows the user accessed a sensitive document at 23:47 and connected a USB drive at 23:49. What hypothesis does this support and what further evidence would you look for?

Practical 3 – Windows Registry Forensic Analysis (Est. time: 4 hrs)

Objective: Extract forensic artefacts from Windows Registry hives to determine program execution history, USB device connections, and persistence mechanisms.

- Environment: Lab machine with Autopsy; same pre-seeded forensic image from Practical 2
- Tools: Autopsy Registry Viewer module (or Registry Explorer if installed)
- Expected output: USB device artefact table (minimum 2 devices); UserAssist decoded program list; Run/RunOnce entries (noting any suspicious entries); RecentDocs list; completed artefact table.
- Observation questions: (1) The USBSTOR key shows a USB drive connected at 23:49 (matching your timeline hypothesis from Practical 2). What additional information from the Registry would confirm data was copied to it? (2) You find a RunOnce entry pointing to C:\Users\Public\svchost32.exe. Why is this suspicious even though the filename looks legitimate?

Practical 4 – Windows Event Log Analysis with Splunk (Est. time: 4 hrs)

Objective: Ingest Windows event logs into Splunk and write SPL queries to identify a brute-force attack, successful compromise, and new account creation.

- Environment: Lab machine with Splunk Free (localhost:8000); faculty-provided Security.evtx and System.evtx
- Tools: Splunk Free

L J INSTITUTE OF COMPUTER APPLICATIONS

Master of Computer Applications (MCA)

Syllabus for Batch - 2025

Semester: III

- Expected output: All 5 SPL query screenshots with results; brute-force timeline chart; 1-paragraph triage note with recommended next step.
- Observation questions: (1) Your query shows 847 failed logins in 3 minutes all from IP 10.0.0.55 targeting the Administrator account. Write the Splunk alert rule that would have triggered on this. (2) Event 4624 Logon Type 3 from a workstation IP using **the Administrator account at 03:00 AM. What does this indicate and why is it suspicious?**

Practical 5 – Network Forensics with Wireshark (Est. time: 4 hrs)

Objective: Analyse a faculty-provided malicious traffic capture to identify attack type, extract credentials, detect beaconing, and build an attack timeline.

- Environment: Lab machine with Wireshark; faculty-provided malicious.pcap
- Tools: Wireshark, NetworkMiner (if installed)
- Expected output: FTP credentials screenshot; SQLi payload in POST body; DNS anomaly screenshot; beaconing pattern screenshot (regular intervals); attack timeline with 6 events.
- Observation questions: (1) The C2 beaconing interval is exactly 60 seconds. How could an attacker make this harder to detect with jitter? (2) The DNS tunnelling used subdomains like dGhpcyBpcyBhIHRLc3Q=.evil.com. What encoding is this and what does the subdomain contain?

Practical 6 – Memory Forensics with Volatility 3 (Est. time: 4 hrs)

Objective: Analyse a Windows memory dump to identify suspicious processes, network connections, and indicators of malware.

- Environment: Kali Linux with Volatility 3; faculty-provided Windows memory dump (.raw) with simulated malware presence
- Tools: Volatility 3 (vol.py), strings, VirusTotal (browser)
- Expected output: Process tree screenshot with suspicious process highlighted; network connections table with anomaly; strings output top 10; VirusTotal result; written analysis paragraph.
- Observation questions: (1) The process tree shows cmd.exe spawned from svchost.exe. Why is this suspicious? What legitimate reasons could cause this? (2) The strings output contains <http://malicious-c2.com/beacon>. What type of malware behaviour does this string suggest?

Practical 7 – Browser & Email Forensics (Est. time: 3 hrs)

Objective: Extract browser forensic artefacts from the forensic image and trace a phishing email to its origin.

- Environment: Lab machine with Autopsy; faculty-provided .eml phishing email file
- Tools: Autopsy Web History module, SQLiteBrowser (optional), text editor
- Expected output: Browser history table; downloads list; email hop-path diagram (drawn); originating IP details; SPF/DKIM analysis; verdict – is the From address spoofed?
- Observation questions: (1) Chrome stores timestamps as WebKit format (microseconds since 1 Jan 1601). Convert 1335000000000000 to a human-readable date. (2) The SPF check result is softfail. What does this mean, and is it sufficient to block the email?

Practical 8 – Splunk Multi-Source Correlation & IR Dashboard (Est. time: 4 hrs)

Objective: Ingest multiple log sources into Splunk and build an IR dashboard that tells the story of a complete attack scenario.

- Environment: Splunk Free; faculty-provided log package (Security.evtx, firewall.csv, web_access.log)

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

- Tools: Splunk Free web UI
- Expected output: Splunk dashboard screenshot showing all 3 panels; SPL queries for each saved report; 1-page incident summary with timeline.
- Observation questions: (1) Your three log sources all record the same attacker IP. How does correlating across sources give you a more complete picture than any one source alone? (2) The web access log shows SQLi from IP 10.5.5.20 but the firewall log shows 10.5.5.20 was blocked 5 minutes earlier. What does this contradiction tell you?

Capstone – Full Forensic Investigation Report (Weeks 11-13, 12 hrs)

Objective: Conduct a complete forensic investigation on a faculty-provided evidence package and produce a legally defensible investigation report.

Evidence package provided by faculty: Pre-seeded Windows disk image + memory dump + network PCAP from a simulated insider data exfiltration incident. The student must determine: what data was accessed, when, how it was exfiltrated, and whether malware was involved.

Deliverable: Case Details & Chain of Custody – Scope & Methodology – Tools Used (with version numbers) – Findings Timeline (chronological, artefact-sourced) – Detailed Findings (each with artefact, location, timestamp, forensic significance) – Conclusion – Recommendations – Appendix (hash values of all evidence items, chain-of-custody form).

Assessment: 30 marks – Artefact identification accuracy (12), Timeline reconstruction quality (8), Report legal defensibility and writing (10).

Reference Books:

1. "Digital Forensics and Incident Response" – Gerard Johansen, Packt Publishing
2. "The Art of Memory Forensics" – Hale Ligh, Case, Levy, Walters, Wiley
3. "Windows Forensic Analysis Toolkit" – Harlan Carvey, Syngress
4. "Incident Response & Computer Forensics" – Luttgens, Pepe, Mandia, McGraw-Hill
5. "Practical Malware Analysis" – Sikorski & Honig, No Starch Press